



CS 412 Intro. to Data Mining

Chapter 3. Data Preprocessing

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Chapter 3: Data Preprocessing

- ❑ Data Preprocessing: An Overview 
- ❑ Data Cleaning
- ❑ Data Integration
- ❑ Data Reduction and Transformation
- ❑ Dimensionality Reduction
- ❑ Summary

What is Data Preprocessing? — Major Tasks

☐ Data cleaning

- ☐ Handle missing data, smooth noisy data, identify or remove outliers, and resolve inconsistencies

☐ Data integration

- ☐ Integration of multiple databases, data cubes, or files

☐ Data reduction

- ☐ Dimensionality reduction
- ☐ Numerosity reduction
- ☐ Data compression

☐ Data transformation and data discretization

- ☐ Normalization
- ☐ Concept hierarchy generation

Why Preprocess the Data? — Data Quality Issues

- ❑ Measures for data quality: A multidimensional view
 - ❑ Accuracy: correct or wrong, accurate or not
 - ❑ Completeness: not recorded, unavailable, ...
 - ❑ Consistency: some modified but some not, dangling, ...
 - ❑ Timeliness: timely update?
 - ❑ Believability: how trustable the data are correct?
 - ❑ Interpretability: how easily the data can be understood?

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Data Cleaning

- ❑ Data in the Real World Is Dirty: Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, and transmission error
- ❑ Incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - ❑ e.g., *Occupation* = “ ” (missing data)
- ❑ Noisy: containing noise, errors, or outliers
 - ❑ e.g., *Salary* = “-10” (an error)
- ❑ Inconsistent: containing discrepancies in codes or names, e.g.,
 - ❑ *Age* = “42”, *Birthday* = “03/07/2010”
 - ❑ Was rating “1, 2, 3”, now rating “A, B, C”
 - ❑ discrepancy between duplicate records
- ❑ Intentional (e.g., *disguised missing* data)
 - ❑ Jan. 1 as everyone’s birthday?

Incomplete (Missing) Data

- ❑ Data is not always available
 - ❑ E.g., many tuples have no recorded value for several attributes, such as customer income in sales data
- ❑ Missing data may be due to
 - ❑ Equipment malfunction
 - ❑ Inconsistent with other recorded data and thus deleted
 - ❑ Data were not entered due to misunderstanding
 - ❑ Certain data may not be considered important at the time of entry
 - ❑ Did not register history or changes of the data
- ❑ Missing data may need to be inferred

How to Handle Missing Data?

- ❑ Ignore the tuple: usually done when class label is missing (when doing classification)—not effective when the % of missing values per attribute varies considerably
- ❑ Fill in the missing value manually: tedious + infeasible?
- ❑ Fill in it automatically with
 - ❑ a global constant : e.g., “unknown”, a new class?!
 - ❑ the attribute mean
 - ❑ the attribute mean for all samples belonging to the same class: smarter
 - ❑ **the most probable value: inference-based such as Bayesian formula or decision tree**

Noisy Data

- ❑ **Noise:** random error or variance in a measured variable
- ❑ **Incorrect attribute values** may be due to
 - ❑ Faulty data collection instruments
 - ❑ Data entry problems
 - ❑ Data transmission problems
 - ❑ Technology limitation
 - ❑ Inconsistency in naming convention
- ❑ **Other data problems**
 - ❑ Duplicate records
 - ❑ Incomplete data
 - ❑ Inconsistent data

How to Handle Noisy Data?

- ❑ Binning
 - ❑ First sort data and partition into (equal-frequency) bins
 - ❑ Then one can **smooth by bin means, smooth by bin median, smooth by bin boundaries**, etc.
- ❑ Regression
 - ❑ Smooth by fitting the data into regression functions
- ❑ Clustering
 - ❑ Detect and remove outliers
- ❑ Semi-supervised: Combined computer and human inspection
 - ❑ Detect suspicious values and check by human (e.g., deal with possible outliers)

Data Cleaning as a Process

❑ Data discrepancy detection

- ❑ Use metadata (e.g., domain, range, dependency, distribution)
- ❑ Check field overloading
- ❑ Check uniqueness rule, consecutive rule and null rule
- ❑ Use commercial tools
 - ▢ Data scrubbing: use simple domain knowledge (e.g., postal code, spell-check) to detect errors and make corrections
 - ▢ Data auditing: by analyzing data to discover rules and relationship to detect violators (e.g., correlation and clustering to find outliers)

❑ Data migration and integration

- ❑ Data migration tools: allow transformations to be specified
- ❑ ETL (Extraction/Transformation/Loading) tools: allow users to specify transformations through a graphical user interface
- ❑ Integration of the two processes
 - ❑ Iterative and interactive (e.g., Potter's Wheels)

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Data Integration

- ❑ Data integration
 - ❑ Combining data from multiple sources into a coherent store
- ❑ Schema integration: e.g., $A.cust-id \equiv B.cust-\#$
 - ❑ Integrate metadata from different sources
- ❑ **Entity identification:**
 - ❑ Identify real world entities from multiple data sources, e.g., Bill Clinton = William Clinton
- ❑ Detecting and resolving data value conflicts
 - ❑ For the same real world entity, attribute values from different sources are different
 - ❑ Possible reasons: different representations, different scales, e.g., metric vs. British units

Handling Redundancy in Data Integration

- ❑ Redundant data occur often when integration of multiple databases
 - ❑ *Object identification*: The same attribute or object may have different names in different databases
 - ❑ *Derivable data*: One attribute may be a “derived” attribute in another table, e.g., annual revenue
- ❑ **Redundant attributes may be able to be detected by *correlation analysis* and *covariance analysis***
- ❑ Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality

Dimensionality Reduction

❑ Curse of dimensionality

- ❑ When dimensionality increases, data becomes increasingly sparse
- ❑ Density and distance between points, which is critical to clustering, outlier analysis, becomes less meaningful
- ❑ The possible combinations of subspaces will grow exponentially

❑ Dimensionality reduction

- ❑ Reducing the number of random variables under consideration, via obtaining a set of principal variables

❑ Advantages of dimensionality reduction

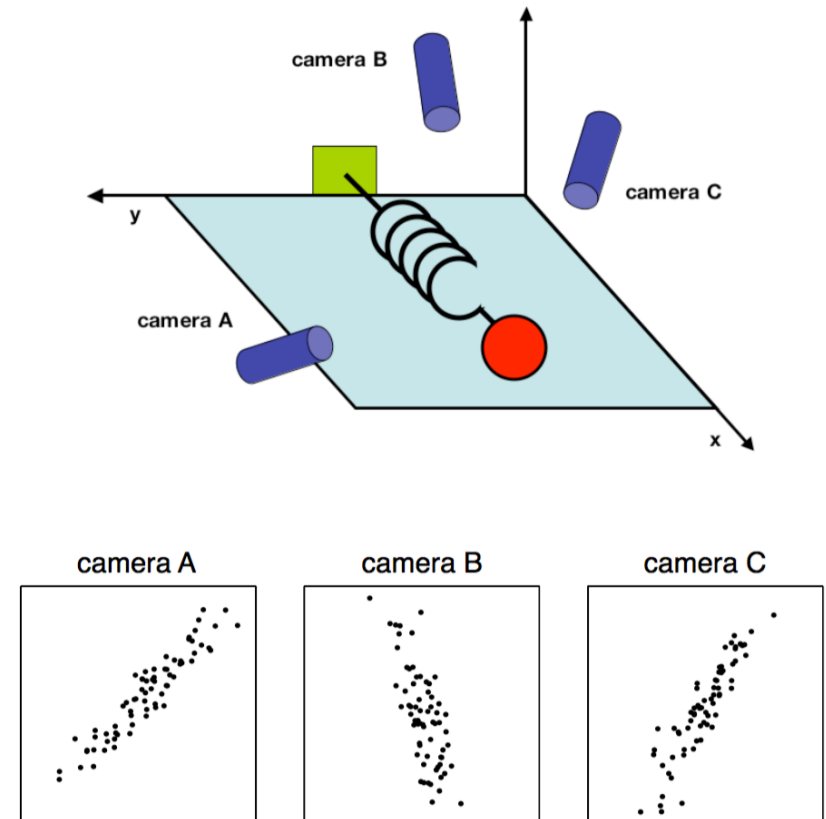
- ❑ Avoid the curse of dimensionality
- ❑ Help eliminate irrelevant features and reduce noise
- ❑ Reduce time and space required in data mining
- ❑ Allow easier visualization

Dimensionality Reduction Techniques

- ❑ Dimensionality reduction methodologies
 - ❑ **Feature selection:** Find a subset of the original variables (or features, attributes)
 - ❑ **Feature extraction:** Transform the data in the high-dimensional space to a space of fewer dimensions
- ❑ Some typical dimensionality methods
 - ❑ Principal Component Analysis
 - ❑ Supervised and nonlinear techniques
 - ❑ Feature subset selection
 - ❑ Feature creation

Principal Component Analysis (PCA)

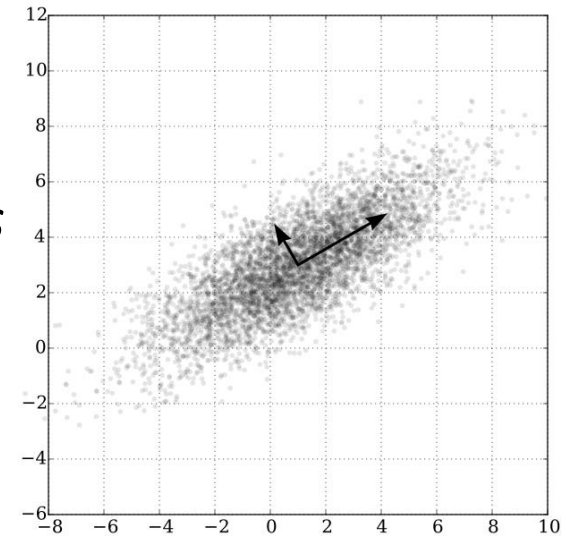
- ❑ PCA: A statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called *principal components*
- ❑ The original data are projected onto a much smaller space, resulting in dimensionality reduction
- ❑ Method: Find the eigenvectors of the covariance matrix, and these eigenvectors define the new space



Ball travels in a straight line. Data from three cameras contain much redundancy

Principal Component Analysis (Method)

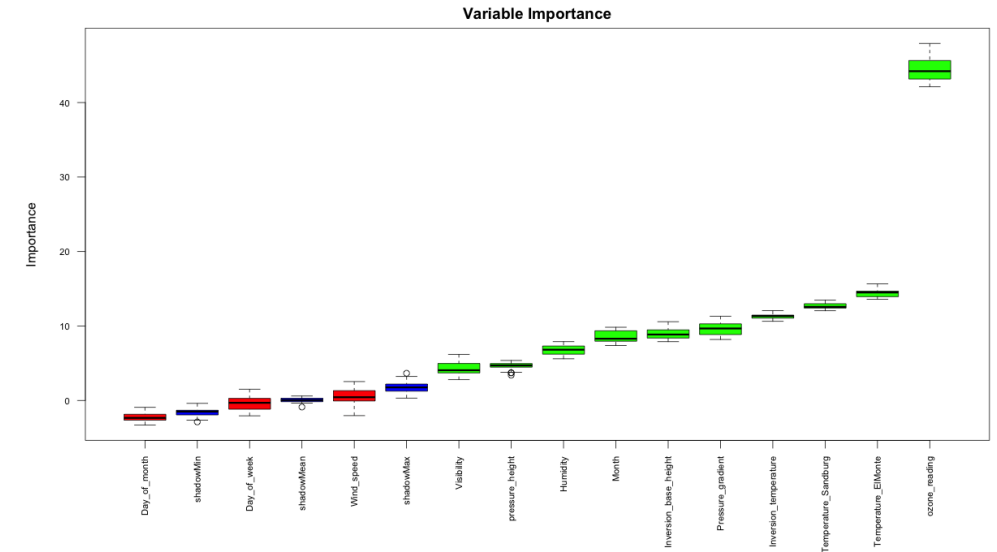
- Given N data vectors from n -dimensions, find $k \leq n$ orthogonal vectors (*principal components*) best used to represent data
 - Normalize input data: Each attribute falls within the same range
 - Compute k orthonormal (unit) vectors, i.e., *principal components*
 - Each input data (vector) is a linear combination of the k principal component vectors
 - The principal components are sorted in order of decreasing “significance” or strength
 - Since the components are sorted, the size of the data can be reduced by eliminating the *weak components*, i.e., those with low variance (i.e., using the strongest principal components, to reconstruct a good approximation of the original data)
- Works for numeric data only



Ack. Wikipedia: Principal Component Analysis

Attribute Subset Selection

- Another way to reduce dimensionality of data
- Redundant attributes
 - Duplicate much or all of the information contained in one or more other attributes
 - E.g., purchase price of a product and the amount of sales tax paid
- Irrelevant attributes
 - Contain no information that is useful for the data mining task at hand
 - Ex. A student's ID is often irrelevant to the task of predicting his/her GPA



Heuristic Search in Attribute Selection

- ❑ There are 2^d possible attribute combinations of d attributes
- ❑ Typical heuristic attribute selection methods:
 - ❑ Best single attribute under the attribute independence assumption: choose by significance tests
 - ❑ Best step-wise feature selection:
 - ❑ The best single-attribute is picked first
 - ❑ Then next best attribute condition to the first, ...
 - ❑ Step-wise attribute elimination:
 - ❑ Repeatedly eliminate the worst attribute
 - ❑ Best combined attribute selection and elimination
 - ❑ Optimal branch and bound:
 - ❑ Use attribute elimination and backtracking

Attribute Creation (Feature Generation)

- ❑ Create new attributes (features) that can capture the important information in a data set more effectively than the original ones
- ❑ Three general methodologies
 - ❑ Attribute extraction
 - ❑ Domain-specific
 - ❑ Mapping data to new space (see: data reduction)
 - ❑ E.g., Fourier transformation, wavelet transformation, manifold approaches (not covered)
 - ❑ Attribute construction
 - ❑ Combining features (see: discriminative frequent patterns in Chapter on “Advanced Classification”)
 - ❑ Data discretization

Summary

- ❑ **Data quality:** accuracy, completeness, consistency, timeliness, believability, interpretability
- ❑ **Data cleaning:** e.g. missing/noisy values, outliers
- ❑ **Data integration** from multiple sources:
 - ❑ Entity identification problem; Remove redundancies; Detect inconsistencies
- ❑ **Data reduction, data transformation and data discretization**
 - ❑ Numerosity reduction; Data compression
 - ❑ Normalization; Concept hierarchy generation
- ❑ **Dimensionality reduction**

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